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BactoMate Custom — user manual

Branch: BactoMate_Custom_New-Analyses

Reviewed: 1 November 2026, against code commit 811d5b6

Upstream baseline/package version: BactoMate 2.9.5 (2dfd181)

This manual describes the current custom application. The earlier V1 and Analyses branches have been removed from GitHub. Upstream release binaries may differ. For a local acceptance checklist use the testing guide; for the complete custom delta use the change summary.

1. Install, launch and update

Use Python 3.11. Clone `santinlab-code/BactoMate` with `--branch BactoMate_Custom_New-Analyses --single-branch`, or download/extract that branch's ZIP. Run commands in the folder containing `main.py`.

macOS:

```
chmod +x setup/macos/*.sh launch_gui.sh
./setup/macos/install.sh
./launch_gui.sh
```

Windows, in a user-writable folder with 64-bit Python 3.11:

```
setup\windows\install.bat
launch_windows.bat
```

The installer creates the project's venv, installs dependencies and the editable package, applies backend patches and performs a smoke check. CPU operation is supported; accelerator availability depends on the backend/environment. Confirm `python main.py --smoke-test` using that venv. A

version string of 2.9.5 alone does not distinguish custom revisions; use `git rev-parse HEAD` with Git.

For a clean Git checkout on the active branch, `git pull --ff-only` updates it. Keep local edits before updating. ZIP users should extract a fresh source folder and create its venv there. Do not move an old venv: launch scripts and editable package paths can still point to the old installation. Never overwrite microscopy results merely to update source. See setup and the Windows guide.

2. Application map and session state

The dashboard and module navigation open File conversion, Segmentation, Tracking, Analysis, Post-processing and Training. Segmentation contains the detection, Filtering and Analyze stages. **Analyze computes measurements; Analysis draws figures from them.** Session Logs and Help are available from the application.

Open a workspace folder to provide a project context. Saved project state uses `bacto-mate_project.json`. Paths already selected in pipeline stages can also supply an Analysis context. Status & Log reports progress and subprocess failures. Save the relevant log when reporting a problem.

Clear current session on the dashboard or File menu closes previews/figures, clears paths, Analysis tables/cache and in-memory assignments, and returns home. Finish or cancel active processing first. It does not delete source images, results, saved projects or saved `analysis_groups.json`. Reopening the same project can therefore restore saved assignments; choose a new workspace for a new experiment.

Session Logs can reload saved processing parameters without replacing current input paths. They are separate from per-object curation, portable filter presets, Analysis assignments and figure recipes.

3. File conversion and XY positions

Select ND2/LIF/CZI input using File conversion. Confirm metadata, calibration and channel roles. Phase is the segmentation background; FI marks a fluorescence measurement channel; Foci marks channels used for spot detection. A channel may carry both fluorescence and foci roles. Names with `_fi_`/`_foci_` help subsequent automatic matching; preserve the common sample and XY identity.

ND2 multiposition conversion processes every XY separately. With Save channels separately, a three-XY/three-channel input gives nine TIFFs. Multiple positions have `XY001`, `XY002`, etc.; single-position data keep conventional names. The selected time series is retained in each stack. Combined-channel output instead produces one multichannel TIFF per XY.

Time selection, Z plane/projection, crop and correction options apply to each XY. A crop rectangle is reused across positions, so check all exported fields. The preview shows the first XY. TIFFs normally go into `Tif_files` beside the source unless an output path is chosen. Pixel/time metadata should be checked before quantitative analysis. Nikon PH3 is recognized as phase. See ND2 details.

4. Cell segmentation

Choose the appropriate Omnipose, Cellpose-SAM or U-Net backend. Use phase/intensity images, not an already rendered colour overlay. Select a built-in or compatible trained model, check device and preview a small field before batch processing.

Control	Interpretation
Model	Backend-specific pretrained or user-trained weights
Diameter	Estimated object size in pixels; auto behavior is backend-specific

Control	Interpretation
Flow / mask thresholds	Model-output acceptance controls; tune against real boundaries
DBSCAN / clustering	Optional backend-specific mask reconstruction
Batch/device options	Throughput and memory use; CPU is available
Overlay/visualization	Saves visual QC images, separate from integer masks

Process one file or a selected folder batch. Preview frame navigation and re-detection help compare settings before a full run. Outputs commonly live in `_segmentation*` directories with mask TIFFs and optional contour overlays. Use the output paths reported in the log rather than inferring them from a screenshot.

Previews show cell outlines and preserve touching-cell boundaries. Show cell outlines toggles them; Outline color changes colour without recomputing masks. Newly exported contour TIFFs use that style. Equivalent-label masks, not these RGB TIFFs, are the inputs to filtering, measurements and tracking.

Detached images and LUTs

Detach image moves the same interactive canvas into a movable window. The LUT histogram follows it and is open on the right. Adjust its Min/Max lines or Auto; changing channels preserves their individual display ranges during navigation. Filtering keys these ranges by mask as well. This viewing memory is not an on-disk LUT preset or a guarantee across application restarts.

Wheel zoom is pointer-centered; hold left-click and drag to pan. Recenter/fit restores the overview, and horizontal/vertical position sliders move the visible center. Escape keeps the detached window open. The parent controls compact at the top while detached; Attach image restores their arrangement.

The close cross asks Save / Discard / Cancel. Save writes the rendered canvas as RGB TIFF/JPEG; it is not a replacement for a label mask or raw intensity TIFF. Cancelling a save keeps the window open. Successful Save/Discard reattaches the canvas. During manual polygon creation, Escape cancels the polygon instead. See segmentation previews.

5. Filtering and manual curation

Open Filtering after detection. The image can remain detached while controls and tables stay in the main interface. Cell filtering/Foci filtering are fixed panels; cell/foci tables and histories provide space for ten rows. **Mask / XY position** above them selects the active sample, mask, detections and curation.

Cell filters and calibrated dimensions

Set pixel size in µm/px, area and length/width limits, aspect ratio, eccentricity, solidity and optional out-of-focus rejection. Check calibration against metadata. A zero/no-limit control should be read using its label/tooltip. New Filtering recipes exclude border-touching cells by default; previously saved choices win.

Area is mask area × pixel-size². Length and width are calibrated major/minor axes of the equivalent ellipse. They are not medial-axis lengths, even when later profiles use a curved skeleton. The table shows these physical dimensions and pixel X/Y centers; invalid calibration does not supply invented physical units.

Apply current filters preserves the View channel. Only retained cells have thin contours, white by default, with a colour selector in the main and detached view. Rejected cells stay in the table for possible reintegration.

Foci channels and filters

Select a foci channel in View; clicking then targets foci in that channel rather than their host cells. Depending on detection data, filters cover probability, intensity/ratio, area and relation to cell boundaries. **Exclude fully outside** is on for new recipes. It is distinct from other outside/touching options.

Kept foci are green and rejected/excluded foci red. Markers derive from saved detections and explicit manual annotations, not arbitrary image maxima created for display. Original detection CSVs keep rejected candidates available. A yellow selection indicator is separate from kept/rejected status.

Select and curate

Input/action	Effect
Left-click object	Select its matching cell/focus table row
Shift-click	Add only clicked objects, without selecting intervening rows
Click background	Clear ordinary object selection
Double-click cell table row	Center image on the cell
Delete/Backspace with image/table focused	Exclude selected active-type objects
Keep selected objects	Force inclusion over automatic filtering
Exclude selected objects	Force exclusion over automatic filtering
Use filters for selected objects	Remove manual overrides

These actions follow the image into its detached window. They are disabled during processing. Exclusion is reversible; it does not erase the source segmentation. Cell and foci decisions are independent, with frame/channel scope as appropriate.

Right-click opens Add / keep object, Remove / exclude object and Create new object here. It never immediately changes kept status just by opening the menu. Existing objects can be reintegrated or excluded. If the target is already in a multiple selection, the relevant action can apply to that selection.

Create a focus at the clicked coordinate in its channel. In Cells mode, the first click starts a polygon; add vertices and finish with Enter/double-click. Escape cancels. The polygon needs at least three pixels and cannot overlap an existing cell. Creations are manual annotations and appear in tables/filtered exports; manual foci have a `manual` marker. Source images/masks remain intact.

Recipes, curation and history

Each XY saves its cell filters, calibration and per-channel foci filters. Visiting a new XY copies the last recipe as a starting point; revisiting an existing XY restores its own values. Channel switching saves/restores values even before Apply. LUT ranges remain independent from scientific filter thresholds.

The mask's `.curation.json` stores manual choices/geometry and local recipes. Keep it with its source mask. A changed segmentation must not receive old object choices against unrelated labels. History records before/after counts, parameters and manual steps for the current frame; it is a record, not an undo command.

Import/Save filter settings above the panels handles portable JSONs: one cells file and one per named foci channel. Importing one channel leaves the others unchanged. Old combined presets remain supported, with validation. Presets do not transfer object-specific selections to unrelated images. See curation and settings.

Save a filtering batch

Filter File/Folder opens the available-mask list with result status, even after previewing one XY. Select the masks to export. Raw and filtered variants of a source are grouped; the raw source is preferred

when present for reversible curation. Each mask uses its own recipe. Outputs include filtered masks, foci CSVs where relevant and portable filter settings. The active preview's recipe is not substituted for all XYs. See batch review.

6. Analyze: quantitative measurements

Select mask files/folder, normally using the saved filtered masks. Check the Analyze-stage calibration and settings; independent Analyze/CLI controls do not necessarily share every Filtering default. Include detailed per-cell data for cell distributions, and enable FI/foci/colocalization outputs needed downstream.

Fluorescence measurement can use one selected FI stack or name-matched channels. Review matching feedback and ensure XY/frame/shape agreement. Contour FI, when enabled, measures an explicitly selected boundary band, not the whole cell. Detection modes include manual/automatic rules and optional Spotiflow. These methods have different candidate/blob definitions; do not treat a displayed peak or manual annotation as evidence of biological localization without review.

When reusing saved segmentation/foci detections, choose the appropriate detection CSV option rather than assuming a fresh independent detector produces identical candidates. Per-channel processing and final merging complete before the Analysis handoff.

Typical workbook roles:

Table	One row represents	Typical content
Summary	Frame	Counts, total/relative area and optional foci summaries
Per-Cell Data	Cell in one frame	ID, morphology, FI, foci counts/aggregates
Foci	Individual focus	Frame/channel, cell link, x/y, intensity and available blob metrics
Colocalization sheets	Matched focus pair	Partner identities, distance and valid/invalid status

Available columns depend on chosen stages/channels. Join foci to cells using source/XY, frame and cell identity; a cell number alone is not globally unique. Combined/master workbooks summarize multiple files and should not be counted again alongside their constituent rows. Also save CSV is off by default; enabling it adds text copies of workbook data. Optional filter visualization is TIFF; automatic PNG companions are disabled.

After successful Analyze and final merges, BactoMate loads the results and opens Analysis automatically, including multi-folder outputs. No manual table import is required for that workflow.

7. Foci distances and colocalization

Open Preview distances from foci analysis to inspect matching before interpretation. Choose mask/XY, frame, two foci channels and cell. New defaults are Max distance 3 px and Threshold 2 px; saved sessions may differ.

Max distance defines candidate partner radius. A one-to-one assignment chooses non-overlapping pairs. Threshold then counts a matched pair as colocalized when its center distance is within the threshold. Same cell only restricts matches when enabled. Raising Max distance alone does not make every pair colocalized. The (?) controls explain the parameters and the preview shares Analyze's logic.

Cyan/magenta mark the channel centers; yellow/green circles show search/threshold radii; green/orange lines indicate valid/above-threshold pairs. Above the selected cell, Colocalized/Not

Colocalized reflects valid pairs involving it. This is an algorithmic distance classification, not proof of molecular interaction.

Checkboxes show phase and multiple fluorescence backgrounds. LUT channel selects the histogram to edit; each channel keeps its contrast. Previous/next-cell buttons and arrow keys navigate cells. Wheel zoom is gentle, and left-drag pans in both views; the unused Home/Pan icons are gone. Calibration, a pixel grid and 1 px reference assist distance reading. Preview edits update analysis settings, but do not redetect foci or write analysis tables. See distance preview.

8. Tracking and identity

Tracking uses Trackastra on a mask time series. Select the masks and optionally the corresponding original TIFF and analysis workbook. Models include general_2d and general_2d_w_SAM2_features. Modes include greedy, greedy_nodiv and ILP; the latter needs compatible solver dependencies from requirements. CPU/CUDA/MPS availability depends on the environment.

Review lineage QC, uncertain links and divisions before exporting tracked masks and lineage data. Persistent CellIDs can be assigned/preserved in supported workflows, but a segmentation label is not automatically a persistent track. The CLI assign_persistent_ids tool can help build IDs; it does not replace visual continuity/division verification. Kymographs use one verified identity and can discover tracked mask exports in `_tracking`.

9. Analysis: results to figures

The separate Analysis module is entirely in English. On entry/Refresh results it uses the workspace or pipeline context to discover result CSV/TSV/XLSX/XLSM. Completed Analyze results are supplied directly after the run. The catalogue caches unchanged files and reports unreadable ones. Known figure exports, CSV/workbook copies and redundant aggregate workbooks are avoided by naming/role rules, not by a universal arbitrary-file deduplication algorithm.

Cells, Foci, Tracking, Profiles and Summary are separate families. Do not use a summary table to represent individual cells. Manual imports remain in Advanced mode for external data. See simple Analysis.

Conditions and replicates

Conditions & replicates stays at the top in simple/advanced modes. Assign each source file a condition and biological replicate, optionally editing selected rows together. Several XYs from one replicate should share a replicate name. Choices are saved in `analysis_groups.json` at the catalogue root. Nonempty assignments override source columns; blank assignments keep source information. The application does not infer biological replication from files or XY numbers.

For an area violin: Cells → Violin plot → Cell area (µm²); choose a frame, Compare conditions and Show replicate means. Each violin represents cell observations; coloured points are replicate means within condition. First-frame defaults avoid unintentional film pooling. All frames explicitly includes repeated observations. No statistical test or hierarchical model is implicitly added.

Figure guide and advanced controls

Figure guide follows four steps: question, population, appearance, review/draw. It finds project masks/channels for spatial figures, prefers filtered masks and reports unfiltered inputs. Spatial computation runs in the background. Advanced / MicrobeJ-style controls keeps manual tables, free axes/groups/filters and profiles. This is inspired by MicrobeJ, not an integration of the Fiji plugin.

All 13 types are available: Histogram, Scatter, Mean curve, Box plot, Violin plot, ECDF, Correlation, Demograph, Kymograph, Normalized cell heatmap, Heatmap, Maxima map and Foci count. The methods reference defines their aggregation and required fields. The figure guide provides recipes.

Spatial profiles and intensity

A demograph has one row per cell observation, sorted by length, with relative longitudinal position 0–1. Principal axis is a straight projection; medial axis uses a pixel skeleton for curved rods and rejects ambiguous/branched axes. Cell-table length/width still use the equivalent ellipse. Geometric alignment does not infer pole age; optional brighter-pole alignment places it at left/top.

Default longitudinal sampling is 24 segments. Native resolution → 24 positions measures about one pixel apart then interpolates onto a common grid; edge values hold the first/last measured mean. `native_profiles.csv` preserves native samples. Neither interpolation nor more segments adds optical resolution. Direct empty bins stay missing; they must not be interpreted as zero fluorescence.

Intensity choice	Formula per cell profile	Main use
Keep absolute fluorescence	I	Compare measured brightness with comparable acquisition/calibration
Scale each cell to its maximum	I / max(I)	Compare distribution shapes while retaining relative baseline
Scale each cell from 0 to 1	(I − min(I)) / (max(I) − min(I))	Compare contrast patterns after removing each profile's minimum

Raw values remain available. A zero denominator maps the normalized profile to zero. These intensity choices do not replace background correction. LUT contrast is independent of this quantitative normalization.

Maxima map distinguishes brightest raw pixel per cell from kept detected foci. Its capsule is schematic and its density is relative point density; neither is a measured cell shape or fluorescence heatmap. Foci count uses per-cell counts, including zeros. Kymographs require one XY/channel and verified identity in at least two frames; time uses minutes when known, otherwise frames.

10. Figure windows, options and export

Each draw creates an independent movable figure window, raised in front of BactoMate with its own data and settings. Show latest figure reopens the latest view. Resizing/exporting a figure does not overwrite another. The live canvas fills its window; export uses a separate off-screen copy.

Figure options includes title/axis labels and limits; category order, display labels and colours; box/violin spacing and width; fonts, X-label rotation, points/opacity, lines/outliers, legend/grid/title; image colour bounds; export size in cm or pixels and raster DPI. Default size is square, 15.24 × 15.24 cm. A normalized-cell schematic has its own appropriate plotting aspect inside that page. Numeric X distances are data; category-spacing controls chiefly apply to boxes/violins. Edit with Figure guide restores that figure's scientific recipe and publication options. Changing the metric clears old axis label overrides.

Escape keeps the window open. The close cross asks Save / Discard / Cancel for an unsaved or changed figure. Save exports the figure/data/code bundle; cancelling its folder selection keeps the figure open. An unchanged successfully exported figure can close directly. Internal session reset closes figures without an individual save prompt: export wanted figures before resetting.

Exports create a new timestamped figure folder containing PDF/SVG, source and plot CSV tables, an Excel workbook, figure settings/provenance, SHA-256 manifest and Python/R/MATLAB scripts. Replicate summaries and native samples appear when used. Automatic PNGs are disabled; an explicitly requested manual PNG is still possible. Scripts redraw saved geometry, not original segmentation; Python execution is tested, while native R/MATLAB execution and renderer/font equivalence need local checks. See methods and exports.

11. Training and post-processing

Training provides a shared dataset workflow plus backend-specific Omnipose, Cellpose-SAM and U-Net controls. Use matching raw image/integer-mask pairs and hold out validation images. Configure tiling, epochs, learning rate and output according to the backend. A compatible user-trained model must still be tested on images outside its training set. Check pipeline and model-inspection tools help diagnose incompatible checkpoint metadata, flat outputs or loading failures. If a custom Omnipose model finds no cells, verify loader patches and thresholds; a more permissive mask threshold can help diagnose it but does not repair poor weights or guarantee a useful model.

Post-processing provides scale bars, montages and growth-curve videos. Scale bars require correct µm/px, length, position, colour and label settings. Montage groups matching channels, normally phase → fluorescence → merge, and uses the first frame; control grid, spacing, labels, colours and LUTs. It is a display product, not a replacement for raw channels.

Growth-curve video combines a TIFF series with Summary data such as Frame and Total Cell Area (Relative). Set minutes per frame, graph limits, playback FPS and TIFF/AVI/GIF/MP4 outputs. Playback speed is distinct from biological time. Relative area is referenced to the first frame and is not an independent cell count.

12. Files, audits and troubleshooting

Artifact	Keep it for
Raw/converted channel TIFF	Source pixels, metadata and matching
Integer segmentation mask	Original detected objects and reversible curation
`.curation.json`	Decisions, manual geometry and local mask/channel recipes
Filtered mask/foci CSV	Selected population used for downstream measurements
`.filter_settings.*.json`	Portable cell or named-channel filters
Analysis workbook/optional CSV	Quantitative rows and summaries
`analysis_groups.json`	Saved condition/replicate assignments
Figure bundle	Figure-specific rows, geometry, settings and reproduction code

File → Inspect generated files is read-only. It compares identical content and known table copies, reports roles and can save JSON; it never deletes files. Raw vs filtered masks and individual vs combined workbooks can be intentional. Automatic pipeline PNG previews are disabled; existing PNGs are not bulk-removed. Documentation PNG illustrations are separate from results.

If Analysis finds nothing, check the current workspace/pipeline context, wait for Analyze to finish and use Refresh results; review read errors. For a missing spatial channel, check common sample/XY names, shapes, calibration and filtered outputs. For an empty demograph, inspect masks, finite intensity and the chosen population. For white/missing strips, inspect NaN bins in plot_data.csv before changing sampling; bright colormap colours can also be near-white.

For Qt startup problems on macOS use setup/macos/repair_pyqt6.sh or recreate the venv. On Windows check Python 3.11 x64, VC++ runtime and the dependency repair commands in the Windows guide. Use the launcher/venv rather than another Python. For model/backend failures run Check environment/Check pipeline and preserve the log. Reduce batch size/use CPU when GPU memory is insufficient.

13. Evidence and limits

The latest recorded full code suite passed 614 tests with 5 skipped; 11 targeted checks passed after the last refinements. Real checks covered a single-time, three-XY/three-channel ND2 with

pixel-identical converted TIFFs, contour review, XY001 curation/export and a 144-cell GFP demograph. Synthetic tests cover further interactions. These results do not certify biological foci detection/colocalization, all real time-lapses, native macOS window behavior or native R/MATLAB execution. The documentation update itself is not a new full application test run.

References

- BactoMate upstream and citation
- MicrobeJ methods
- Omnipose
- Cellpose
- Trackastra
- Documentation index, CLI reference
- Release notes, paper data